Discussion

Interpretation boundary I

This pipeline describes the shape of a retrieved literature slice: its phase coverage, metadata completeness, topic structure, citation links, and declared evidence. It does not adjudicate scientific truth. Hypothesis scores, when available, summarize recorded assertions and should be read as evidence-landscape signals rather than calibrated probabilities.

Retrieval and phase bias I

Engine availability, query wording, rate limits, temporal boundaries, identifier quality, and full-text access all shape the retained corpus. The retrieval report is the authoritative account of source outcomes; counts must never be reverse-engineered from the merged corpus. Phase overlap and cross-phase citation rates are descriptive checks, not proof that phases are independent or exhaustive.

Fixture and live modes I

The offline path uses reserved synthetic identifiers and generated records. Fixture outputs validate the machinery and its contracts only. A live review must replace the fixture, retain source-level provenance, refresh the claim ledger, and obtain domain review before reporting substantive findings. The manuscript therefore uses explicit pending and fixture-derived states rather than silently promoting demonstration values.

Limitations and extensions I

- Keyword taxonomies can misclassify ambiguous records; changes belong in configuration and should be accompanied by a regenerated classification report.
- TF-IDF/SVD embeddings are lexical and deterministic; transformer-based alternatives require an explicit dependency and evaluation decision.
- LLM extraction is optional, model-sensitive, and not a substitute for source review.
- Full-text availability is a coverage measure, not a quality measure.
- Reproducibility claims require a clean double run and normalized artifact comparison.

These limitations are represented as method-stage and review-required findings so a publication decision can distinguish deterministic failures from editorial judgment.